

ASHUTOSH PRADHAN

Software Engineer · Distributed Systems · AI-Driven Solutions

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SUMMARY

Results-driven Software Developer with hands-on experience building scalable, high-performance applications. Focused on deepening expertise in **distributed systems** and **AI-driven solutions**, with the long-term goal of becoming a skilled software architect who designs resilient systems that solve complex real-world problems and create meaningful impact.

WORK EXPERIENCE

Innovate Technologies — Kathmandu, Nepal

Apr 2026 – Present

Software Engineer

- Built **RESTful** APIs in **Express.js** and **NestJS** over **PostgreSQL** with **JWT/bcrypt** auth, role-based authorization, and **Zod**-validated, **Swagger**-documented endpoints, delivering secure CRUD services with transaction-safe bulk operations.
- Designed a memory-efficient **Excel ingestion pipeline** using the **NestJS Stream API** with a producer/consumer architecture, backpressure handling, and batched **ACID transactions**, processing large files without memory exhaustion or partial data writes.
- Implemented distributed rate limiting via a custom **NestJS ThrottlerGuard** backed by shared **Redis**, throttling by user ID with IP fallback and fail-open handling, enforcing consistent limits across instances while degrading gracefully during cache outages.
- Integrated **REST** APIs into the frontend with dynamic routing and auth flows, using **TanStack Query** for cached fetching and **Zustand** for persistent state, reducing redundant network calls and improving perceived load times.
- Adopted **gRPC** and **GraphQL** for internal service-to-service communication within a **monorepo** architecture, standardizing type-safe contracts and simplifying cross-team integration.
- Refactored legacy state management from **Redux Saga** to **Redux Toolkit**, simplifying async logic and reducing boilerplate for a more maintainable frontend codebase.
- Optimized document viewing performance by replacing traditional **PDF rendering** with a virtualized React PDF viewer using **react-pdf** and **react-virtuoso**, enabling efficient rendering of large documents.
- Built a custom email template management system with scheduled **CRON** jobs, enabling configurable emails to be triggered based on user actions and events such as subscription activity.

FiveOne — Kathmandu, Nepal

Aug 2025 – Apr 2026

Software Engineer (Part-time)

- Migrated the database from **MongoDB** to **PostgreSQL** within a modular **NestJS** backend, improving relational data integrity and query performance for a real-time social platform.
- Built core features of the platform's **React Native** mobile application, implementing cross-platform UI components and API integrations for a consistent experience across web and mobile.
- Migrated real-time **WebSocket** event handling from the client into the **NestJS** backend, centralizing AI events, game-state updates, and chat synchronization for more reliable multiplayer sessions.
- Implemented AI-powered conversational features using the **OpenAI LLM API** with **prompt engineering**, increasing per-session user engagement by **~25–35%** and reducing early chat drop-offs.
- Integrated **Firebase** (Auth, Analytics, Messaging) with **Google/Apple Sign-In** and **Notifee** for secure login and reliable push-notification engagement.

Gunasys — Kathmandu, Nepal

Apr 2024 – Aug 2025

Software Engineer

- Implemented secure authentication and **role-based access control (RBAC)** to manage permissions across multiple business roles and departments.
- Developed and optimized **RESTful** backend services and database schemas, integrating **in-memory caching** to reduce average transaction latency to **<150ms**.
- Improved frontend performance through state-management optimization, caching strategies, and efficient API handling for smoother user experiences.
- Built **LLM-powered RAG pipelines** to deliver a domain-specific medical chatbot supporting one-on-one patient queries.

Islington College — Kathmandu, Nepal

Feb 2025 – Aug 2025

Academic Liaison Officer

- Conducted structured student follow-ups on attendance and coursework submissions, driving a **35% reduction** in non-submissions and a measurable increase in overall attendance.
- Managed academic staff scheduling and workload allocation, achieving **zero class cancellations** throughout tenure and significantly improving timetable efficiency.

Fresh KTM — Kathmandu, Nepal

Oct 2023 – Apr 2024

Frontend Intern

- Designed and deployed user-centric workflows with dynamic dashboards and real-time updates, improving inventory tracking and order-management efficiency by **~25%**.
- Collaborated with senior developers to enhance FreshKTM's **ERP** web app, implementing responsive frontend components and state-management optimizations.

PROJECTS

KaKhani — Restaurant Reviews with NLP — kakhani.netlify.app

FastAPI, Next.js, Python, PostgreSQL, Jupyter

- Designed and developed a full-stack restaurant review platform enabling users to submit, browse, filter, and rate restaurants with paginated, searchable listings.
- Implemented **RESTful** backend services using **FastAPI**, following a clean layered architecture with DTOs, service abstraction, and repository patterns.
- Built an **NLP**-based review-analysis pipeline that extracts sentiment scores and keywords, feeding aggregate restaurant ratings and sentiment-weighted rankings.
- Integrated **Redis** caching with timed and action-based invalidation, and secured the API with **JWT**-based authentication and role-based authorization.

Anyani — Animated Image Generator — anyani.netlify.app

Python, Next.js, Jupyter Notebook

- Built an animated face-generation pipeline using a **ResNet-18** feature extractor combined with **Conv2D** layers for facial-attribute encoding.
- Implemented a **GAN** architecture (Generator + Discriminator) in **TensorFlow / PyTorch** to synthesize animated face images.
- Optimized the model with **CUDA**, batch normalization, and residual connections to improve performance and convergence.
- Deployed the generator behind a **FastAPI** backend with a **Next.js** frontend for on-demand animated-face generation.

EDUCATION

Islington College

Oct 2021 – Dec 2024

BSc (Hons) Computer Science and Information Technology — First Class Honours

Rato Bangla School

Jun 2018 – Oct 2020

SKILLS

- **Languages:** JavaScript/TypeScript, Python, Java, GoLang, C#
- **Frameworks:** NestJS, Express.js, FastAPI, React.js, React Native, Next.js, Gqlgen
- **Tools:** PostgreSQL, MongoDB, Redis, Docker, AWS, GraphQL, gRPC, Git, Firebase

SOFT SKILLS

Adaptability · Problem Solving · Communication · Management · Teamwork

CERTIFICATIONS

- **AWS Academy Cloud Foundations** — 2022
- **AWS Academy Machine Learning Foundations** — 2022

AWARDS

- **AAA Scholarship**, Islington College — July 2025

REFERENCES

Available on request